

Performance Benchmarking of Embedded Battery State Estimation: A Measurement-Only Comparison of Direct-Measurement, Hybrid, and Data-Driven Estimators

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Abstract

This paper presents a measurement-only robustness benchmark for battery State-of-Charge (SOC) estimation under realistic Battery Management System constraints. Four estimator classes are compared: direct measurement by Coulomb counting (DM), SOH-adaptive Coulomb counting (HDM), a second-order equivalent-circuit model with extended Kalman filtering (HECM), and a GRU-based data-driven estimator (DD). Neural-model development and final evaluation are cell-disjoint, while the preidentified HECM parameterization is held fixed. The final benchmark covers 16 predeclared 24-h windows from six independent Lithium Iron Phosphate holdout cells, 19 nominal and disturbed-measurement cases, and 6,720 causal model runs. Randomized scenarios use fixed repeated seeds; cells, rather than time samples, are treated as independent units in a hierarchical bootstrap with 10,000 repetitions. Under nominal measurements, DD provides the lowest cell-macro error (MAE 0.0258, RMSE 0.0300), followed by HECM, HDM, and DM. The adverse signed current-gain penalty grows monotonically for all estimators; at 3%, the MAE increase is 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD. HECM

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corrects an imposed 10-percentage-point initialization mismatch fastest on average, whereas DD combines comparatively strong initialization and post-dropout recovery. Missing samples and burst dropout are more consequential than timing jitter, and voltage spikes reveal a local DD transient that is largely hidden by full-window MAE. A paired reference-SOH ablation shows that replacing the causal LSTM-SOH trace does not uniformly improve SOC error, emphasizing that the estimators were calibrated for their deployed feature pipeline. Finally, isolated SOC-core tests on an STM32H753ZI verify close software–hardware agreement and quantify latency and static memory. The results support scenario-specific model selection while limiting conclusions to the tested LFP cells, operating envelope, models, and implementation protocol.

Keywords: Battery management system, State of charge, State of health, Performance benchmark, Direct measurement, Equivalent circuit model, Data-driven estimation, Embedded artificial intelligence

1 Nomenclature

2 *Acronyms*

3	BMS	Battery management system
4	SOC	State of charge
5	SOH	State of health
6	DM	Direct-measurement model
7	HDM	Hybrid direct-measurement model
8	HECM	Hybrid equivalent-circuit model
9	DD	Data-driven neural model
10	ECM	Equivalent circuit model
11	EKF	Extended Kalman filter
12	MAE	Mean absolute error
13	RMSE	Root-mean-square error
14	ADC	Analog-to-digital converter

15	CV	Constant-voltage phase
16	OCV	Open-circuit voltage
17	GRU	Gated recurrent unit
18	LSTM	Long short-term memory
19	LFP	Lithium iron phosphate
20	<i>Symbols</i>	
21	t	Time
22	$I(t)$	Current signal at time t
23	$U(t)$	Voltage signal at time t
24	$T(t)$	Temperature signal at time t
25	$\widehat{\text{SOC}}(t)$	Estimated state of charge at time t
26	$\widehat{\text{SOH}}(t)$	Estimated state of health at time t
27	$\text{SOC}_{\text{ref}}(t)$	Reference SOC target at time t
28	$\text{SOH}_{\text{ref}}(t)$	Reference SOH target at time t
29	ΔI	Current bias or offset
30	ΔU	Voltage bias or offset
31	ΔT	Temperature bias or offset
32	σ_I	Standard deviation of injected current noise
33	σ_U	Standard deviation of injected voltage noise
34	σ_T	Standard deviation of injected temperature noise
35	Q_c	Online cumulative charge state used by the SOC models
36	C_{nom}	Nominal capacity
37	C_{eff}	Effective capacity used inside the estimator
38	U_{max}	Upper voltage threshold used for full-charge detection
39	U_{tol}	Voltage tolerance used in the full-charge condition
40	U_1, U_2	Polarization voltages of the first and second RC branch
41	ΔMAE	Error increase relative to baseline

42 1. Introduction

43 1.1. Motivation and practical relevance

44 Reliable battery state estimation is a core BMS function because SOC and
45 SOH determine safety margins, usable energy, operating limits, charge con-
46 trol, and degradation-aware system management in both mobile and stationary
47 lithium-ion battery applications [1, 2, 3]. Practical deployment quality is not de-
48 fined by clean-condition accuracy alone, because real BMS operation is exposed
49 to sensor bias, additive noise, initialization uncertainty after restart, missing
50 samples, irregular sampling, communication interruptions, and transient signal
51 spikes caused by measurement-chain limitations or disturbances in the surround-
52 ing system [4, 5, 6]. The relevant engineering question is therefore not only
53 whether an estimator achieves low MAE under nominal measurements, but also
54 whether it converges back to a reliable state after disturbed initialization and
55 whether it remains stable under physically plausible measurement corruption.

56 As summarized conceptually in Figure 1, the present benchmark treats
57 deployment-facing performance as a three-dimensional problem consisting of
58 nominal accuracy, convergence behavior after state inconsistency, and robust-
59 ness against disturbed measurements and signal-integrity faults, embedded within
60 broader BMS requirements such as hardware constraints and implementation
61 effort. The present study therefore isolates the performance branch and com-
62 pares representative estimator classes under identical measurement-side distur-
63 bances and a matched embedded-feasibility constraint. The benchmark shows
64 that strong nominal accuracy does not necessarily imply strong disturbance
65 performance and that the estimator classes respond differently to bias, missing
66 samples, and re-initialization events.

67 1.2. State of the art in battery state estimation

68 The literature commonly groups battery-state estimators into direct-measurement
69 or integration-based approaches, model-based approaches based on equivalent-
70 circuit or electrochemical models, hybrid approaches that combine physical

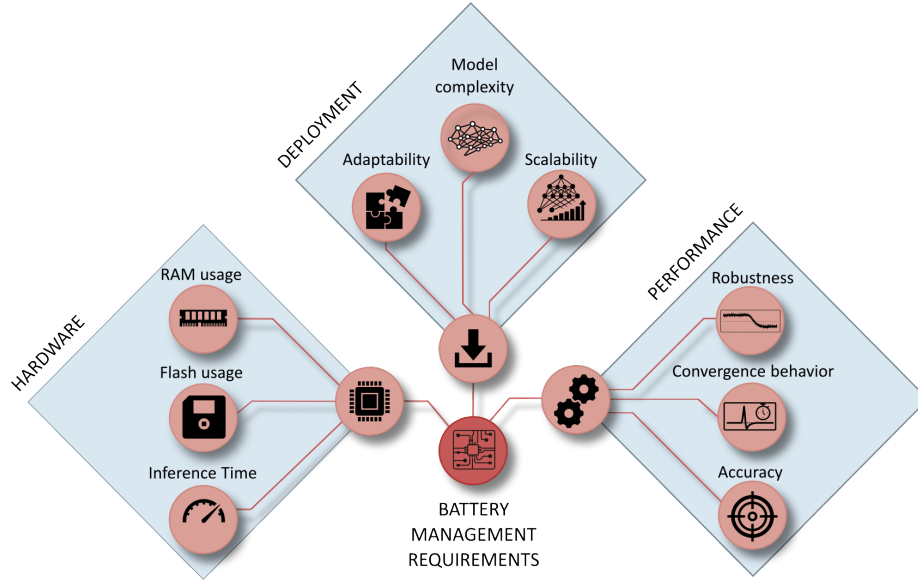


Figure 1: Conceptual overview of deployment-oriented requirements for embedded BMS state estimation. The broader requirement space includes implementation effort, hardware constraints, and estimator performance, whereas the present study focuses on the performance branch and evaluates baseline accuracy, convergence behavior, and robustness under measurement disturbances.

71 structure with learned components, and purely data-driven models based on
 72 neural sequence learning [7, 1, 2, 3]. Direct-measurement approaches, typically
 73 based on Coulomb counting, are attractive because they are simple, computa-
 74 tionally inexpensive, and easy to deploy; however, they remain structurally sen-
 75 sitive to current bias, missing data, and initialization errors because they lack an
 76 internal correction mechanism [7, 4]. Model-based approaches, especially ECM-
 77 and observer-based estimators, offer physical interpretability and voltage-based
 78 correction capability, but their practical behavior depends strongly on the qual-
 79 ity of model parametrization, operating-point observability, and measurement
 80 quality [8, 2, 4].

81 Hybrid estimators seek to combine physical state propagation with learned
 82 adaptation of capacity, parameters, or auxiliary states, which has made joint
 83 SOC/SOH estimation an active research direction [9, 10, 11]. This combination

can improve baseline calibration, while it does not necessarily eliminate structural sensitivity to measurement disturbances. Data-driven approaches based on recurrent or sequence models can exploit nonlinear cross-relations between current, voltage, temperature, and derived online features, and recent work has also demonstrated their embedded feasibility through TinyML-oriented implementations, pruning, and quantization [12, 13, 14, 15, 3].

1.3. Performance and embedded deployment gap

A large share of the SOC estimation literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, while comparatively fewer studies analyze how estimators behave under realistic measurement faults and timing corruption [1, 13, 14, 15]. Existing robustness studies often remain confined to a single model family, for example observer-based robustness against modeling and measurement errors [4] or robust estimation concepts for specific BMS architectures under noise and uncertainty [5], whereas cross-family comparisons under the same disturbance protocol and under comparable embedded constraints remain limited. Benchmarking studies in the broader BMS context underline the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-level performance benchmarking across fundamentally different estimator classes [6]. This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence, robustness under disturbed measurements, and computational constraints.

1.4. Contribution and study objectives

This work addresses the identified gap through four contributions. First, it compares four deployment-oriented estimator classes under one causal protocol on 16 predeclared windows from six cell-disjoint LFP holdouts rather than on one selected trajectory. Second, it applies 19 physically interpretable measurement, timing, availability, quantization, and initialization cases with repeated

113 stochastic seeds. Third, it uses one estimator-independent recovery criterion and
 114 cell-level hierarchical inference instead of treating time samples as independent
 115 replicates. Fourth, paired reference-SOH runs isolate the shared SOH-service
 116 contribution, while STM32 measurements quantify numerical equivalence, la-
 117 tency, and static memory of the SOC cores.

118 The manuscript should therefore be read as a controlled cross-class esti-
 119 mator benchmark, not as a complete system-level BMS validation campaign
 120 or a universal ranking of estimator families. It isolates estimator-side failure
 121 modes under matched data, disturbance definitions, online execution rules, and
 122 evaluation metrics, and translates the resulting evidence into scenario-specific
 123 guidance for embedded BMS estimator selection.

124 **2. Methodology**

125 *2.1. Estimator classes under test*

126 Figure 2 summarizes the causal benchmark chain, from measured input sig-
 127 nals and online feature reconstruction to disturbance injection and the final
 128 global and local performance evaluation. The direct-measurement model (DM)
 129 represents the canonical embedded Coulomb-counting approach with fixed nom-
 130 inal capacity and explicit full-charge reset logic, and serves as the low-complexity
 131 reference for integration-driven SOC estimation. All SOH-dependent estima-
 132 tor variants in the benchmark, namely HDM, HECM, and DD, use the same
 133 shared LSTM-based online SOH estimator; the classes differ only in how this
 134 common SOH signal is consumed inside the SOC pipeline. The hybrid direct-
 135 measurement model (HDM) preserves the same Coulomb-counting core but re-
 136 places the fixed capacity assumption with an online effective capacity derived
 137 from this shared SOH estimator, which makes it representative of lightweight
 138 hybrid estimation. The hybrid equivalent-circuit model (HECM) represents
 139 physics-based state estimation with observer correction, because it combines
 140 current-driven state propagation with voltage-residual feedback and SOH-dependent
 141 parameterization in a second-order RC equivalent-circuit structure driven by the

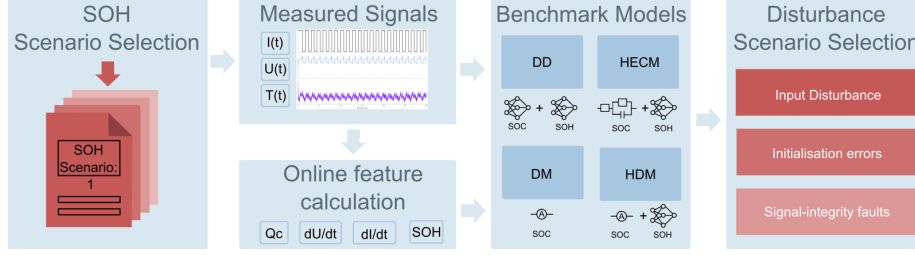


Figure 2: Methodology overview for the causal benchmark chain. Measured battery signals are converted into online auxiliary features, processed by representative estimator classes, exposed to measurement-only disturbances, and finally evaluated through global and local performance metrics.

142 same shared SOH estimate [8, 2]. The data-driven model (DD) represents a neu-
 143 ral estimator pipeline in which SOC is predicted by a lightweight rolling-window
 144 GRU sequence model while the shared LSTM-based online SOH estimate is pro-
 145 vided as causal context under the same execution constraints [12, 3].

146 2.2. SOC estimation methods

147 2.2.1. Direct-measurement model (DM)

The direct-measurement SOC formulation used in DM, HDM, and as an explicit auxiliary feature in DD follows the same online cumulative-charge state, denoted here uniformly as Q_c . It is updated causally from the measured current stream as

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (1)$$

where I_k is given in ampere and Δt_k in seconds, so that the factor 3600 s h^{-1} converts the increment to ampere-hours. The sign convention is fixed by the benchmark implementation and the state is reset to $Q_c = 0$ after a sustained constant-voltage full-charge event. In the DM, this same charge state is mapped directly to SOC through the fixed nominal capacity

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right), \quad (2)$$

148 *2.2.2. Hybrid direct-measurement model (HDM)*

The HDM preserves exactly the same Coulomb-counting core and differs only in the capacity term. The SOH estimate rescales the usable capacity as

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (3)$$

which yields

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (4)$$

149 This formulation makes the relation between DM and HDM explicit: both use
 150 the same online charge state Q_c , while only the denominator changes through
 151 online SOH support. In both direct-measurement variants, a sustained near-
 152 maximum voltage condition is interpreted as a constant-voltage full-charge event;
 153 if the voltage remains above $U_{\text{max}} - U_{\text{tol}}$, where U_{tol} denotes the admissible volt-
 154 age margin below the upper threshold U_{max} , for the configured CV duration, the
 155 internal charge state is reset so that $Q_c = 0$ and therefore $\widehat{\text{SOC}} = 1$, which re-
 156 flects the practical assumption that a completed CV phase provides a physically
 157 anchored full-charge reference.

158 *2.2.3. Hybrid equivalent-circuit model (HECM)*

The hybrid ECM model is implemented as a second-order RC model with the state vector

$$\mathbf{x}_k = \begin{bmatrix} \widehat{\text{SOC}}_k & U_{1,k} & U_{2,k} \end{bmatrix}^\top, \quad (5)$$

where U_1 and U_2 denote the polarization voltages of the two RC branches. The prediction step propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1 (1 - e^{-\Delta t_k/\tau_1}) \\ R_2 (1 - e^{-\Delta t_k/\tau_2}) \end{bmatrix}, \quad (7)$$

where $C_b = C_0 \widehat{\text{SOH}}$ is the SOH-scaled available capacity. The corrected terminal-voltage estimate is written as

$$\widehat{U}_k = \text{OCV}(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k) + U_{1,k} + U_{2,k} + R_i I_k, \quad (8)$$

and the Kalman update uses the residual between $\widehat{U}_k$ and the measured terminal voltage to correct the internal state. In the present implementation, the state propagation is driven by the previous current sample I_{k-1} , whereas the measurement equation uses the current sample I_k available at the correction step. The ECM parameters are not constant. The implementation interpolates R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC from a preidentified lookup table over SOC and SOH, with separate charge and discharge parameter surfaces. The HECM is implemented as a two-RC observer with SOH-dependent parameter scheduling.

2.2.4. Data-driven SOC model (DD)

The data-driven SOC model is a GRU-based recurrent estimator with the feature vector $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{SOH}, Q_c, dU/dt, dI/dt, \Delta t\}$. The benchmarked structurally pruned and fine-tuned checkpoint consists of a single-layer GRU with hidden size 67 followed by an MLP head with one hidden layer of size 96 and a sigmoid output layer. During optimization, the recurrent core is exposed to causal sequence segments of length $L_{\text{SOC}} = 2024$ samples. Within this feature set, Q_c is exactly the same online cumulative-charge state defined in (1). It is included explicitly so that the neural estimator receives the same charge-throughput information that also drives DM and HDM. The derivative channels are computed online as the finite differences $dI/dt(k) = (I_k - I_{k-1})/\Delta t_k$ and $dU/dt(k) = (U_k - U_{k-1})/\Delta t_k$. They provide local transient information that is not captured by the absolute channels alone. The additional feature Δt_k allows the network to distinguish amplitude changes from sampling-interval changes under irregular timing. During the primary benchmark, each output is calculated from the latest causal 2024-sample window and recurrent state is reset between windows, matching training and validation. No future information is

used. Continuous hidden-state forwarding is evaluated only as a separate deployment ablation because it changes the learned inference semantics.

2.3. SOH estimation method

The shared SOH estimator used in the current benchmark is an LSTM-based sequence-to-sequence model that operates on hourly aggregated online features derived from $\{U \text{ [V]}, I \text{ [A]}, T \text{ [}^\circ\text{C]}, \text{EFC}, Q_c\}$, where the aggregation uses the statistics mean, standard deviation, minimum, and maximum for each feature [16, 17, 18, 19]. This transforms the five base channels into a 20-dimensional hourly feature vector. Architecturally, the SOH network first projects the aggregated input through a two-layer feature embedding block with size 128, layer normalization, GELU activations, and dropout, then processes the sequence with a two-layer unidirectional LSTM of hidden size 112, and finally maps the recurrent output through three residual MLP blocks and a prediction head with hidden size 160 to obtain the SOH estimate for each hourly step.

During optimization, the SOH estimator is exposed to causal hourly sequence segments, and benchmark execution processes one hourly aggregate at a time with causal hidden- and cell-state forwarding. This allows the estimator to accumulate degradation context without access to future information; the first hourly output is anchored with the configured initial SOH value before subsequent hourly predictions are taken directly from the recurrent model output. The SOH prediction is updated at hourly cadence and then held piecewise constant between update points, which enforces an online deployment constraint and avoids sample-wise access to future information while still allowing the faster SOC estimators to use the latest SOH context. This cadence matches the slowly varying capacity fade observed in the dataset but cannot represent an abrupt capacity-loss event, which is absent from the measurements and outside the study claims. In the HDM, the predicted SOH enters the SOC computation through the effective capacity relation $C_{\text{eff}} = C_{\text{nom}} \cdot \widehat{\text{SOH}}$, so the method remains structurally close to direct measurement while replacing the fixed-capacity assumption. In the HECM, the predicted SOH affects the capacity scaling and

215 the lookup-based ECM parametrization, while in the DD model it is provided
216 as an explicit SOC input feature that contextualizes the current measurement
217 trajectory.

218 *2.4. Model and target-hardware preparation*

219 The software benchmark uses frozen floating-point checkpoints and scalers
220 selected before holdout evaluation. Neural training, validation, hyperparameter
221 selection, scaler fitting, pruning, and checkpoint selection use only the declared
222 training and validation cells; no holdout cell enters these steps. The imported
223 HECM lookup surfaces were preidentified from separate HPPC experiments and
224 held fixed before this benchmark; no scored trajectory or benchmark result was
225 used to fit or retune them. The compact SOC-GRU has hidden size 67 and
226 the shared SOH-LSTM has hidden size 112. Configuration, checkpoint, and
227 scaler hashes are stored with the campaign manifest, and the runner rejects
228 cells outside the declared holdout split unless an explicit diagnostic override is
229 supplied.

230 The target-hardware experiment evaluates the isolated SOC cores on a NUCLEO-
231 H753ZI board with a 480 MHz STM32H753 Cortex-M7. To separate SOC im-
232 plementation cost from the shared SOH service, the board receives the precom-
233 puted causal SOH trace used by the software reference; the SOH-LSTM itself
234 is not executed on the microcontroller. The hardware results therefore measure
235 numerical equivalence, latency, and memory of the SOC implementations rather
236 than end-to-end SOC-SOH system latency. Device time is measured per valid
237 inference with the data-watchpoint-and-trace cycle counter. Each cell sequence
238 is replayed three times, firmware section sizes are extracted from the release
239 ELF images, and predictions are compared with both the software implemen-
240 tation and dataset SOC. Reported RAM is static ELF allocation including the
241 linker-reserved heap/stack block, not measured peak stack or dynamic activa-
242 tion memory.

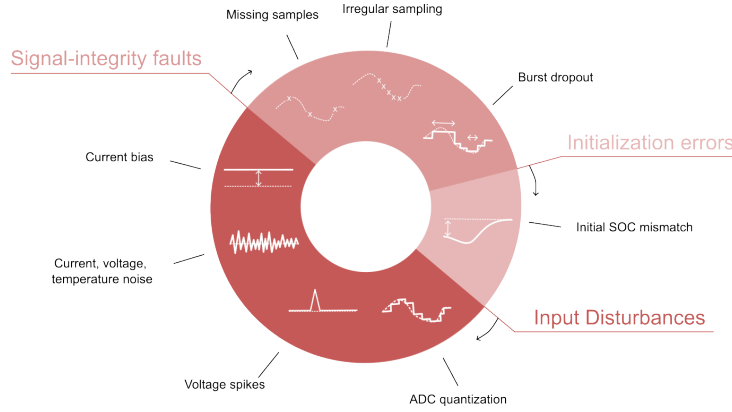


Figure 3: Taxonomy of the disturbance scenarios used in the performance benchmark, grouped into input disturbances, initialization errors, and signal-integrity faults.

2.5. Performance benchmark design

The benchmark follows a strict measurement-only principle: only quantities that a real BMS measures or derives online from measured data are manipulated, namely current, voltage, temperature, sample availability, and sampling time, whereas artificial parameter mismatch and synthetic hidden-state corruption are deliberately excluded [6]. Figure 3 groups the benchmark into input disturbances, initialization errors, and signal-integrity faults, while Table 1 defines the exact disturbance implementations used in the study.

The disturbance magnitudes are not intended as one-to-one copies of a single field specification. Instead, they were selected from three justification classes: deployment-level sensing limitations, adverse measurement corruption, and deliberate signal-integrity stress tests [20, 21, 22, 23, 24, 25]. The main benchmark matrix follows the scenario families shown in Figure 3 and provides the basis of the main result figures and tables. ADC quantization is part of the same scenario set and is reported in Figure 16. Overall, the benchmark is designed as a controlled and reproducible scenario study that exposes estimator-specific failure modes under a common disturbance protocol rather than emulating the complete system stack of a production BMS.

Table 1: Disturbance configurations used in the benchmark and their practical rationale. Exact numeric values are fixed benchmark settings; the cited sources motivate disturbance type and magnitude range.

Scenario	Benchmark setup	Practical origin / rationale
ADC quantization	$\Delta I = 0.01 \text{ A}$; $\Delta U = 0.005 \text{ V}$; $\Delta T = 0.5 \text{ }^\circ\text{C}$	ADC steps; sensor scaling; rounding; coarse deployment-level discretization [20, 21, 26]
Current gain error	Signed gain sweep: $I_{\text{meas}} = I(1 + b)$ with $b = \pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$	Sensor gain error; miscalibration; drift; upper bound as stress case [21, 22, 23]
Current noise	Gaussian; $\sigma_I = 0.02 \text{ A}$ and 0.10 A	Sensing-chain noise; EMI pickup; low and high sensor-noise levels [4, 5]
Voltage noise	Gaussian; $\sigma_U = 0.01 \text{ V}$	Wiring noise; front-end noise; poor filtering; EMI [4, 20, 24]
Temperature noise	Gaussian; $\sigma_T = 1.0 \text{ }^\circ\text{C}$	Sensor tolerance; conversion noise; thermal gradients; sensor-to-cell mismatch [26, 11]
Initial SOC error	Initial mismatch: -0.10 SOC ; DD via perturbed online Q_c	Restart; storage; missing recent history; weak LFP OCV anchoring [4, 8]
Missing samples	Every 50th sample frozen; separate random 2% loss	Packet loss; blocked updates; logger dropouts; periodic and random data freeze [5, 25]
Irregular sampling	Uniform jitter: $\pm 0.1 \text{ s}$, $\pm 0.5 \text{ s}$, and $\pm 0.9 \text{ s}$ around nominal 1 s grid	Scheduling jitter; bus latency; asynchronous acquisition [6, 5, 25]
Burst dropout	One central frozen gap: 3600 s	Communication interruption; frozen task; stuck data stream; severe recovery case [6, 5]
Voltage spikes	Single-sample spikes: $\pm 0.20 \text{ V}$ every 1000 samples (1000 s)	EMI; switching disturbance; corrupted voltage samples [24]

2.6. Metrics and evaluation criteria

For deployment-oriented interpretation, the benchmark results are synthesized along three performance dimensions: nominal accuracy under baseline conditions, convergence behavior after initialization mismatch or interrupted signal continuity, and robustness under sustained or transient measurement corruption. Disturbance-related global performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error, because these metrics provide a compact view of average performance, systematic deviation, and tail behavior across the selected evaluation windows. Local disturbance performance is analyzed through scenario-specific metrics such as jump

counts, output variance, absolute-error variance, and excess rolling MAE, because several disturbances do not primarily manifest as window-wide MAE collapse but as short-term output fluctuations, delayed drift, or slow re-synchronization. Whenever a disturbance mask exists, the analysis further separates pre-disturbance, disturbed, and post-disturbance behavior in order to quantify disturbance-window error growth, recovery time, and residual post-disturbance error.

ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted primarily through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally use dedicated local evidence because their most relevant effects are episodic or recovery-related. Recovery comparisons use one estimator-independent criterion: absolute SOC error below 0.02 for a continuous 300 s, evaluated up to a common 24 h censoring horizon. Runs that do not recover are retained as censored observations rather than silently omitted. The comparison reports recovery-or-censoring time, the censored fraction, initial post-event error, MAE after 1 h and 6 h, and excess-error area under the curve. The requested -0.10 initialization mismatch is mapped into each estimator’s available online state: the coulomb-counting SOC initialization for DM and HDM, the EKF SOC state for HECM, and the capacity-equivalent Q_c offset for DD. Because the DD mapping passes through the recurrent network rather than directly fixing its output, every realized initial output error is reported before recovery is compared. The resulting ranking compares deployment-accessible initialization interventions, not mathematically identical latent-state perturbations.

The independent statistical unit is the holdout cell, not an individual time sample or selected window. Predeclared SOH windows are first averaged within each cell and random seed; the six cells then receive equal macro weight. Hierarchical bootstrap resampling with seeds nested inside cells and 10,000 repetitions provides 95% confidence intervals. Scenario effects are evaluated as paired, cell-matched differences from baseline, and estimator pairs are compared using exact two-sided sign-flip tests, raw mean and median differences, paired standardized

effect sizes, and Holm-adjusted p -values. Because only six independent cells are available, effect magnitudes and confidence intervals are treated as primary evidence and p -values as supporting evidence.

3. Experimental Setup

3.1. Dataset and data acquisition

The benchmark uses the MGFarm LFP 18650 dataset because its controlled design-of-experiments aging study provides long 1-Hz full-cell trajectories, repeated capacity checkups, multiple operating conditions, and enough independent cells for a cell-disjoint comparison [27, 28, 29]. Model development and final testing use a fixed split. SOC and SOH training use C01, C03, C05, C11, C17, and C23; validation uses C07, C19, and C21; and the independent holdout benchmark uses C09, C13, C15, C25, C27, and C29. The six test cells span distinct aging, thermal, duration, and load envelopes, as summarized in Figure A.21. LFP’s flat mid-SOC OCV and chemistry-specific aging behavior are integral to the tested problem; conclusions therefore do not transfer directly to NMC/NCA or pack-level operation.

The holdout cells are assigned to descriptive load groups before model results are inspected, using visible separation in the measured 95th-percentile absolute C-rate: C25 and C27 form the low-load group, C09, C13, and C15 the middle-load group, and C29 the high-load case. Because the high-load group contains one cell, its group-by-SOH results are explicitly exploratory and are not interpreted as population-level high-load inference. Within each trajectory, performance is additionally stratified by reference SOH (fresh: $\text{SOH} \geq 0.90$; mid-life: $0.80 \leq \text{SOH} < 0.90$; aged: $\text{SOH} < 0.80$), instantaneous absolute C-rate (< 0.5 , $0.5\text{--}1.5$, and ≥ 1.5), temperature ($\leq 30^\circ\text{C}$, $30\text{--}35^\circ\text{C}$, and $> 35^\circ\text{C}$), and SOC (< 0.20 , $0.20\text{--}0.80$, and > 0.80). Absent states are reported as not available and are never imputed.

3.2. Reference targets and preprocessing

The reference preprocessing first computes the signed transferred charge as $Q_k = I_k \Delta t_k / 3600$ and the absolute transferred charge as $|Q_k|$, from which the capacity during dedicated capacity-test intervals is obtained as $C_{\text{ref}} = \sum_k |Q_k|$. The reference SOH target is then defined offline as $\text{SOH}_{\text{ref}} = C_{\text{ref}} / C_{\text{ref},0}$, where $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the resulting capacity and SOH columns are interpolated between the dedicated capacity-test anchors. The reference SOC target used in the benchmark corresponds to the offline Coulomb-counting target $\text{SOC}_{\text{ref}} = 1 + Q_c / C_{\text{ref},0}$, where the cumulative charge state Q_c is propagated from the signed charge and reset to 0 at the upper voltage anchor and to $-C_{\text{ref},0}$ at the lower voltage anchor during preprocessing. These reference targets are intentionally treated as offline evaluation quantities, because they rely on dataset-side processing and anchor logic that are not fully available to an online estimator during deployment.

3.3. Test trajectory and benchmark scenarios

All estimators are evaluated on every available holdout-cell/SOH combination under the same disturbance protocol. One representative 24-h window is frozen for each available fresh, mid-life, and aged state, yielding 16 windows; C27 contributes only a fresh window because its measured SOH does not fall below 0.90. Windows start at a measured full-charge anchor ($\text{SOC} \geq 0.98$, $U \geq 3.58 \text{ V}$) and are selected as measured-feature medoids without using estimator outputs or errors. The one-hour dropout uses 48 h from the same anchor to retain approximately 24 h of post-gap observation. The shared SOH LSTM receives 192 h of undisturbed causal context before every scored window, matching its trained sequence length. Figure A.23 summarizes this protocol.

The 19 predeclared cases include the nominal baseline, two current-noise levels, voltage and temperature noise, three current-bias levels, voltage and temperature offsets, ADC quantization, initial SOC mismatch, periodic and random sample loss, three timing-jitter levels, a one-hour burst dropout, and

randomized-sign voltage spikes. Gaussian sensor-noise cases use 10 deterministic seeds; random sample loss, jitter, and spike cases use 5 seeds; deterministic cases run once. Figure A.22 maps each case to the manipulated measurement or state and marks the selected paired reference-SOH ablations. Input disturbances are applied directly to measured channels, initialization errors only where structurally equivalent, and signal-integrity scenarios to sample availability or timing while preserving causal estimator execution.

3.4. Implementation details

Every estimator is executed causally, and all auxiliary quantities required by the models are rebuilt directly from disturbed inputs. This applies to Q_c , EFC, dU/dt , dI/dt , and hourly SOH aggregates, so no hidden dataset-side feature leaks into the estimator input. The SOH LSTM forwards hidden and cell states across hourly updates after its 192-h initialization context. The primary SOC-GRU evaluation preserves its trained sequence-to-one semantics: each prediction uses the latest 2024-sample rolling window with recurrent state reset between windows. A faster continuous-stateful stream is treated only as a deployment ablation because it changed full-trajectory DD accuracy and therefore cannot silently replace the trained benchmark semantics. In freeze and dropout scenarios, disturbed or frozen measurements propagate consistently into derived online features. Direct-measurement estimators use the same CV-triggered reset that defines the full-charge start, whereas the DD initial-SOC test uses an equivalent offset of online Q_c because the network exposes no explicit SOC state. The environment enforces identical capacity assumptions, disturbed streams, targets, and metrics across all estimators.

4. Results

4.1. Baseline performance

Figure 4 summarizes nominal performance across the 16 frozen windows. Equal weighting of the six holdout cells yields a different ranking from the

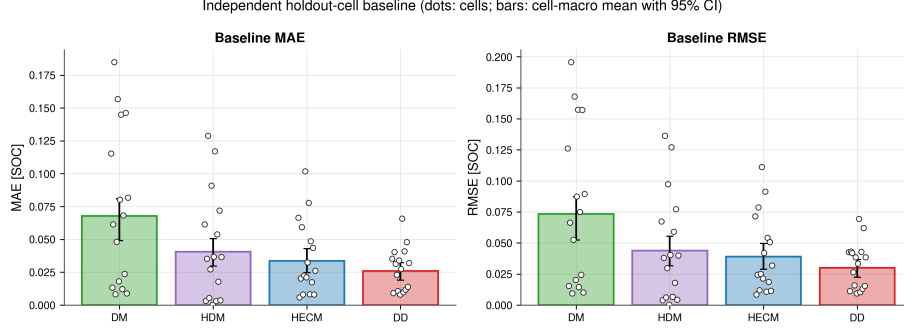


Figure 4: Baseline performance across six independent holdout cells. Dots show cell/SOH-window values; bars and error bars show the equal-weight cell-macro mean and hierarchical 95% confidence interval. DD has the lowest macro MAE and RMSE, while DM has the largest error and between-cell spread.

previous single-trajectory analysis: DD has the lowest baseline MAE at 0.0258 (95% CI 0.0189–0.0323), followed by HECM at 0.0335 (0.0246–0.0429), HDM at 0.0405 (0.0297–0.0506), and DM at 0.0678 (0.0491–0.0809). The RMSE ranking is identical, with values of 0.0300, 0.0389, 0.0438, and 0.0733, respectively. Cell-level points reveal substantial between-cell variation, particularly for DM, so the bars must be interpreted as cell-macro summaries rather than performance on a homogeneous trajectory.

4.2. Current-bias sensitivity

Figure 5 summarizes the adverse response to a signed current-gain sweep at $\pm 0.5\%$, $\pm 1.5\%$, and $\pm 3.0\%$. For each magnitude and holdout cell, the larger MAE change from the two signs is reported because the opposite sign can partially compensate a pre-existing baseline error. The resulting worst-case degradation increases monotonically with gain-error magnitude for every estimator. DM and HDM show the strongest response, HECM is less sensitive, and DD is least sensitive. At 3%, the cell-macro MAE increases by 0.0107 for DM, 0.0100 for HDM, 0.0059 for HECM, and 0.0038 for DD.

The continuous C29 lifecycle analysis in Figure 6 separates three mechanisms that independent 24-h windows cannot resolve. The bias contribution varies over

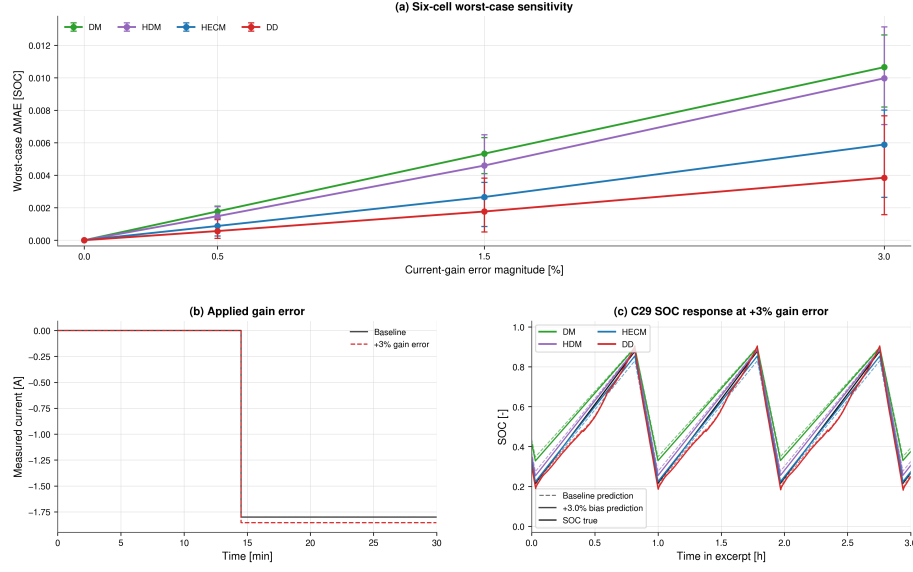


Figure 5: Current-gain sensitivity. (a) Six-cell adverse-direction ΔMAE for each matched $\pm b$ pair; error bars are hierarchical 95% confidence intervals. (b) Representative current before and after the +3% gain error. (c) C29 SOC trajectories over three charge–discharge intervals. The adverse-pair summary prevents signed compensation of pre-existing baseline bias from being misread as improved robustness.

life because its signed Coulomb-counting error interacts with charge direction
and operating state. Between full charges, the penalty generally grows within
an interval, but full-charge anchoring can partially reduce or restructure it. The
process is therefore neither an unrestricted monotonic drift nor a complete reset.
Across the full replay, DD retains the smallest adverse penalty, whereas DM and
HDM remain most exposed to accumulated current error.

4.3. Noise sensitivity

Repeated sensor noise is weaker than persistent current gain error, but the
response depends on channel and estimator structure. At $\sigma_I = 0.10$ A, HECM
has the largest macro penalty ($\Delta\text{MAE} = 0.0056$), whereas DM changes by
 -0.0009 and HDM and DD remain near zero. Small negative changes are paired
finite-window effects, not evidence that noise improves an estimator in general.

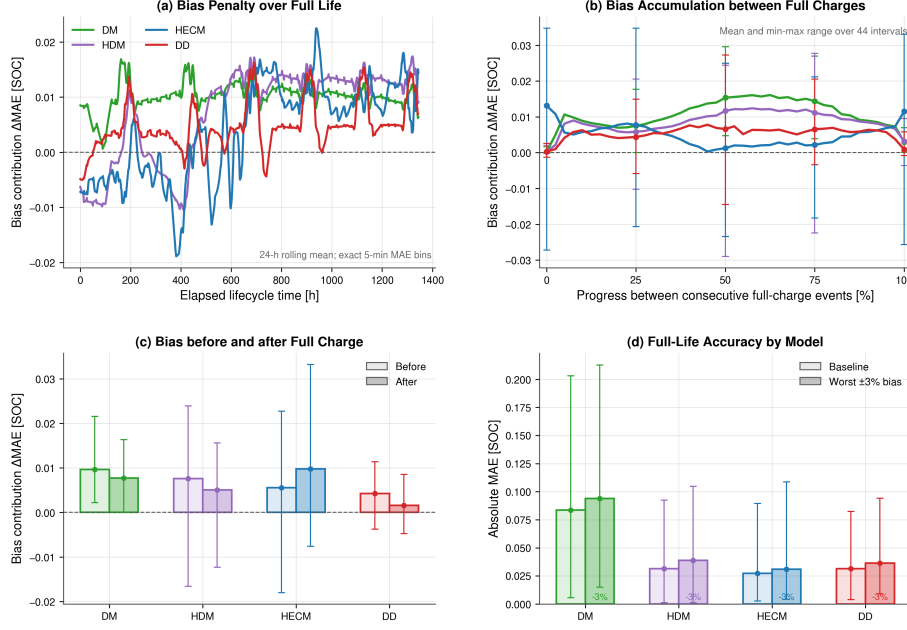


Figure 6: Current-bias accumulation over the continuous C29 life replay. (a) Rolling 24-h bias contribution over elapsed life. (b) Mean and range over normalized intervals between consecutive full-charge events. (c) Penalty immediately before and after full-charge events. (d) Full-life baseline and adverse-bias MAE. Full-charge events reduce or restructure accumulated error but do not erase the lifetime penalty uniformly.

416 Voltage noise ($\sigma_U = 0.01$ V) produces a 0.0014 penalty for DM and 0.0013
 417 for HDM, while HECM and DD remain below 0.0003. Temperature noise is
 418 neutral for DM and small for the SOH-aware models. Figure 7 shows the local
 419 transmission of current noise, while Figure 8 reports repeated six-cell effects
 420 with hierarchical uncertainty.

4.4. Initialization error and recovery

422 Initialization performance is evaluated using one model-independent target
 423 criterion: the shifted output must remain within 0.02 SOC of the paired correct-
 424 initialization output for at least 300 s, with censoring at 24 h. The same re-

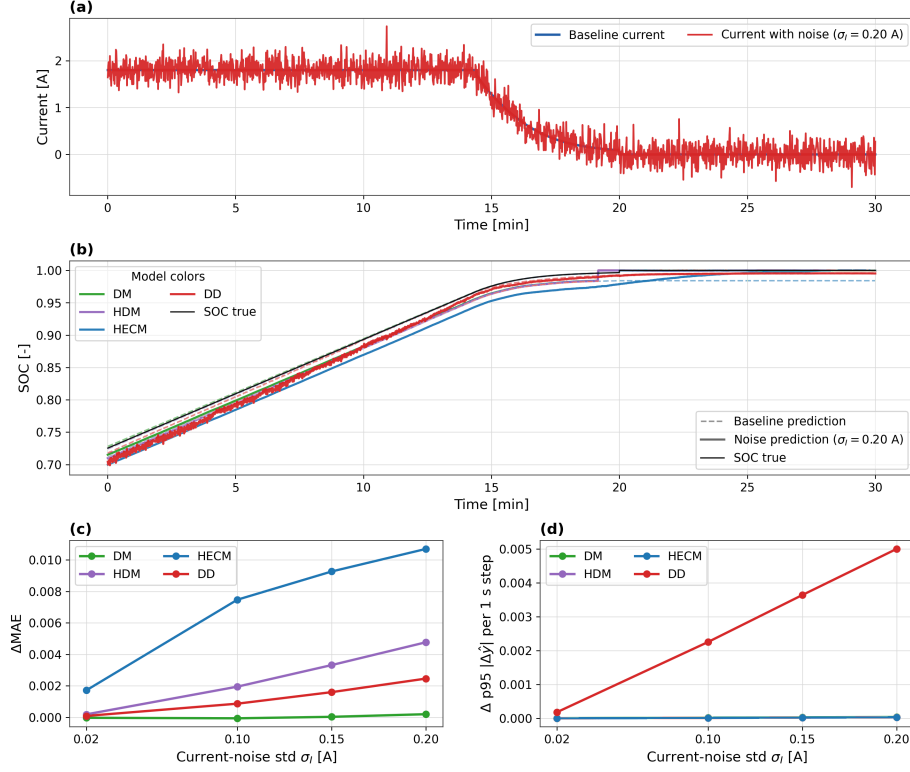


Figure 7: Noise sensitivity under additive current noise. The figure combines an applied current-noise example, a matched SOC comparison in the same 30 min window, the global MAE penalty, and the local output-variability sensitivity across noise levels.

425 requested -0.10 SOC-equivalent mismatch is mapped to the explicit DM/HDM
 426 Coulomb-counting state, the HECM EKF SOC state, and a Q_c offset of $-0.10C_{\text{nom}}$
 427 for DD. Figure 9 reports the realized shift rather than assuming that these in-
 428 ternal mappings create identical outputs. Across cells, HECM recovers fastest
 429 at 1.03 h on average, followed by DD at 5.57 h; DM and HDM both average
 430 22.28 h and are frequently close to the censor horizon. Thus, observer correction
 431 is most effective for the imposed initialization mismatch, whereas integration-
 432 based states generally require a later physical anchor.

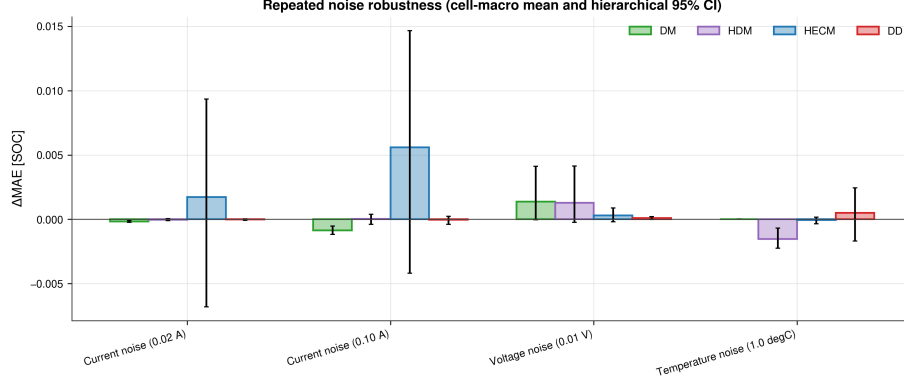


Figure 8: Repeated noise robustness across six holdout cells. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel-model combinations remain close to baseline relative to the between-cell uncertainty.

4.5. Missing samples and signal integrity

Sample loss is substantially more consequential than timing jitter. Periodic freezing of every 50th sample increases MAE by 0.0071, 0.0064, 0.0028, and 0.0011 for DM, HDM, HECM, and DD, respectively; random 2% loss produces corresponding increases of 0.0079, 0.0080, 0.0038, and 0.0026. By contrast, even the ± 0.9 s jitter case remains within approximately 6×10^{-4} SOC of baseline for every model, with confidence intervals generally spanning zero. DD therefore has the smallest sample-loss penalty, while DM and HDM retain their structural dependence on uninterrupted charge integration.

The one-hour burst dropout is a distinct recovery problem. Its macro ΔMAE is largest for HECM (0.0221), followed by HDM (0.0134), DD (0.0134), and DM (0.0067). After measurements resume, DD has the shortest mean recovery-or-censor time (4.13 h), then DM (5.13 h), HDM (5.67 h), and HECM (12.45 h). The hours-long response has model-specific origins: DM/HDM retain the unobserved charge deficit until a physical anchor, HECM must correct it through weak LFP voltage observability, and DD recovers as its rolling input context is replaced by valid samples. Figure 11 therefore reports the local transition in

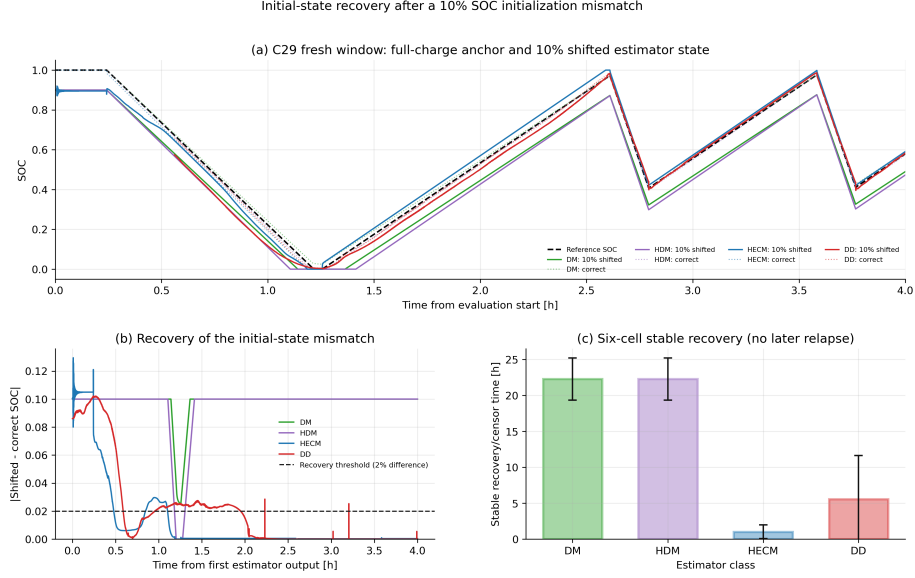


Figure 9: Recovery after a 10% SOC-equivalent initialization mismatch. (a) Representative paired trajectories. (b) Difference between shifted and correctly initialized outputs with the common 2% threshold. (c) Stable recovery-or-censor time across the six holdout cells with hierarchical 95% confidence intervals.

450 addition to the global score.

451 4.6. Spike sensitivity

452 Single voltage spikes demonstrate why local and global metrics must be re-
 453 ported together. Full-window ΔMAE remains below 9×10^{-4} SOC for every
 454 estimator and is effectively zero for DM, HDM, and DD. Event-aligned analysis,
 455 however, identifies a reproducible DD transient: its mean spike-sample penalty
 456 across cells is approximately 7.7×10^{-3} SOC, compared with near-zero direct
 457 penalties for the other classes. Figures 12 and 13 further show that the DD re-
 458 sponse varies across cells and SOH states and then decays after the event. This
 459 pattern is consistent with a one-sample voltage impulse entering both the abso-
 460 lute voltage and dU/dt channels and perturbing the rolling recurrent prediction.
 461 It is a mechanism-supported interpretation, not proof of a unique causal input
 462 channel because no channel-removal ablation was performed.

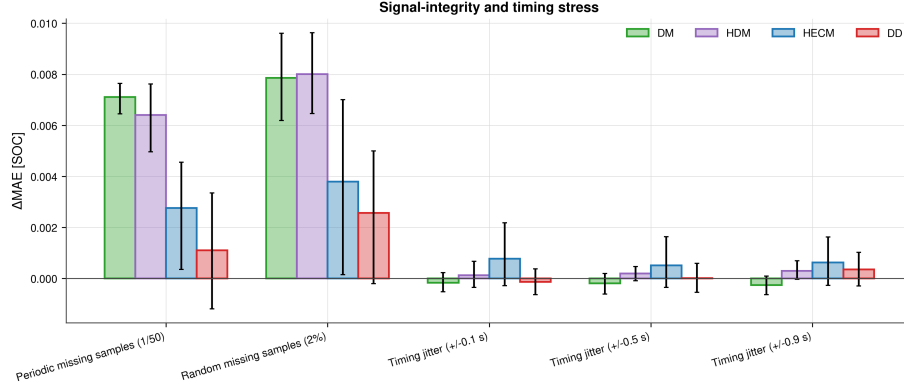


Figure 10: Six-cell signal-integrity and timing stress. Bars show cell-macro ΔMAE and hierarchical 95% confidence intervals for periodic and random sample loss and three timing-jitter levels. Sample loss clearly dominates timing jitter under the tested protocol.

4.7. Shared-SOH ablation

The primary comparison intentionally supplies one common causal LSTM-SOH trace to HDM, HECM, and DD. A paired ablation repeats selected scenarios with dataset reference SOH to quantify this shared-service confounder. Reference SOH reduces baseline MAE by 0.0363 for HDM and 0.0164 for HECM, but increases DD MAE by 0.0049 on average. The same direction persists for most noise, sample-loss, and dropout cases. At 3% current gain error, reference SOH still improves HDM and HECM but worsens DD by 0.0091. For temperature noise, the reference-minus-LSTM difference changes from its baseline value by only 0.0015 for HDM and less than 10^{-4} for HECM and DD, indicating that the incremental temperature-noise response is not dominated by the shared SOH service. This does not imply that a less accurate SOH estimate is intrinsically beneficial; rather, the models consume SOH differently and were calibrated with different feature interactions. The LSTM-SOH primary results therefore represent deployable pipeline performance, while reference-SOH runs are explanatory ablations rather than replacement rankings.

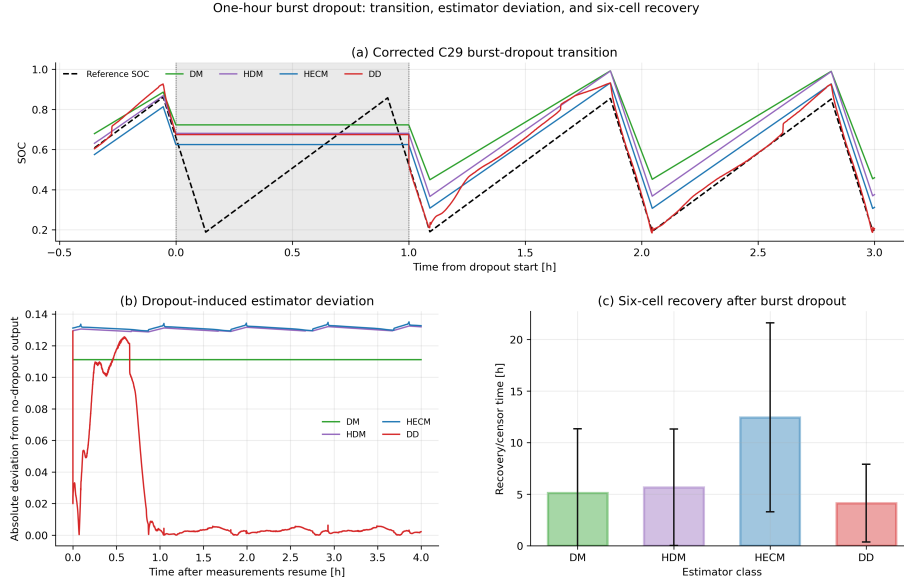


Figure 11: One-hour burst-dropout transition. (a) Baseline outputs (dashed) and dropout outputs (solid) around the shaded missing-input interval. (b) Absolute deviation from the paired no-dropout output after measurements resume. (c) Six-cell recovery-or-censor time under the common 2%/300-s criterion.

4.8. Cross-scenario comparison

Figure 14 condenses the complete measurement-only campaign. Persistent and missing-information faults dominate the global penalties: current-gain error rises with magnitude for all models, periodic/random sample loss particularly affects DM and HDM, and the one-hour dropout produces the largest penalty for HECM. By contrast, the three timing-jitter levels and voltage spikes remain near zero in full-window MAE, although the spike analysis above reveals a DD-local transient. DD has the smallest adverse current-bias penalty and lowest sample-loss penalties; HECM has the largest high-current-noise and dropout penalties but the fastest initialization recovery. The final data therefore favor DD in the aggregate robustness score without implying uniform dominance in every individual mechanism.

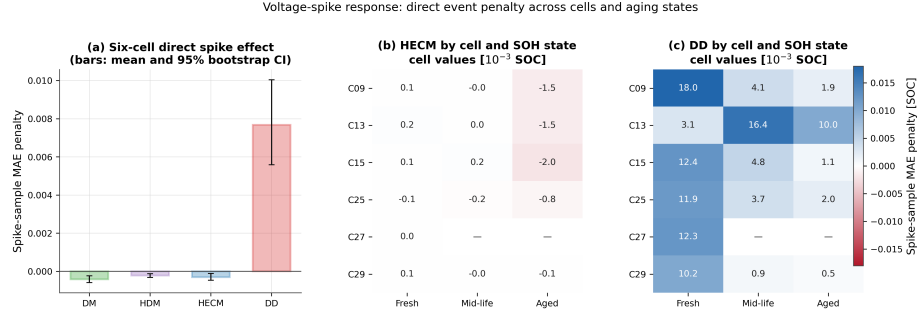


Figure 12: Voltage-spike sensitivity across six cells. (a) Mean event-sample MAE penalty with hierarchical 95% confidence intervals. (b,c) Cell- and SOH-state-resolved penalties for HECM and DD, respectively; absent cell/state combinations are left blank. The event-aligned metric exposes the local DD response that is obscured in full-window Δ MAE.

4.9. ADC quantization

The tested ADC steps ($\Delta I = 0.01$ A, $\Delta U = 0.005$ V, and $\Delta T = 0.5$ °C) preserve the local signal shape but affect the estimator classes differently. DM and HDM increase by 0.0037 and 0.0034 MAE, respectively. HECM changes by -0.0018 and DD by $+0.0002$; both confidence intervals overlap zero. Thus, quantization at these deliberately coarse steps is secondary to current gain error, sample loss, and burst dropout in this dataset, but it is not negligible for the integration-based models.

4.10. Target-hardware validation

The STM32 experiment first verifies numerical equivalence before comparing speed or memory. Across the six nominal cell sequences, mean hardware–software MAE remains below 4×10^{-4} percentage points for every SOC core and reaches 3.41×10^{-6} percentage points for DD (Figure 17). These differences are orders of magnitude below the estimator-to-dataset errors and show that the float32 C implementations reproduce their software references sufficiently closely for deployment measurements.

The isolated DM and HDM SOC updates both have sub-microsecond median latency (0.171 and 0.165 μ s), HECM requires 23.9 μ s, and continuous-state DD

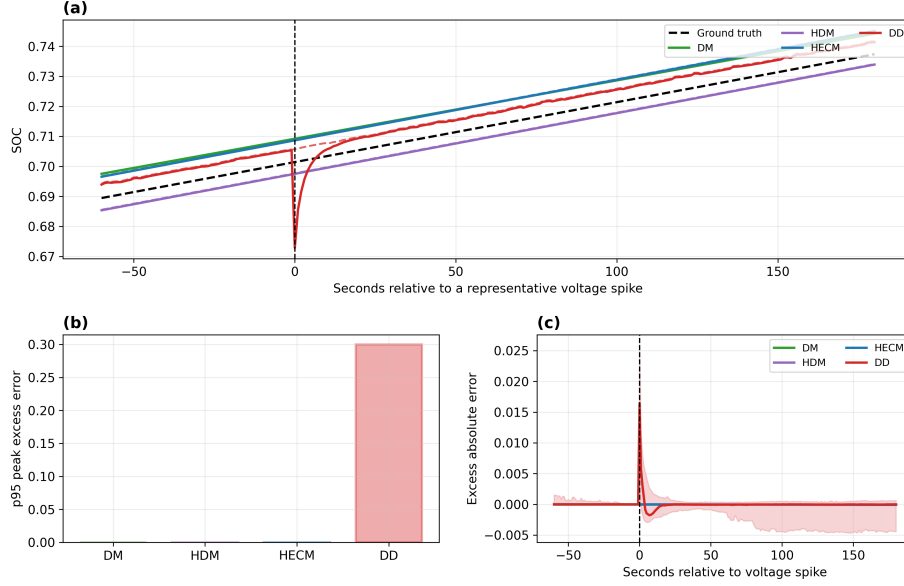


Figure 13: Event-aligned voltage-spike response. (a) Representative SOC trajectories around the injected spike; the dashed black line is reference SOC. (b) Model-wise 95th percentile peak excess error. (c) Mean excess absolute error aligned across repeated spikes, with the DD uncertainty envelope. The local response decays rapidly and is therefore weakly represented by full-window MAE.

requires 424.9 μs (Figure 18). At the nominal 1-Hz update rate, these medians occupy less than 0.043% of one sampling period, although this isolated deadline share is not a measurement of total BMS CPU load. The DD value in this cross-model comparison is the continuous-state deployment variant; exact 2024-sample rolling-window semantics are evaluated separately because recomputing the full sequence at every sample is much slower.

Release images occupy 27.0, 27.1, 470.1, and 116.1 KiB flash for DM, HDM, HECM, and DD, respectively. Static RAM is approximately 4.8 KiB for DM/HDM, 4.9 KiB for HECM, and 69.8 KiB for rolling-window DD (Figure 19). HECM flash is dominated by its parameter surfaces, while the DD rolling window dominates static RAM. These values describe the isolated firmware images and do not include the shared SOH-LSTM service; static RAM also remains a lower

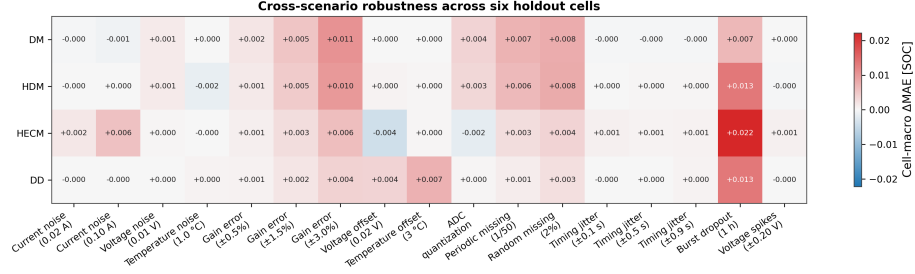


Figure 14: Cell-macro Δ MAE across the complete six-cell measurement-only campaign. Positive values denote degradation relative to each model’s paired baseline; small negative values reflect finite-window signed effects and are not interpreted as beneficial disturbances.

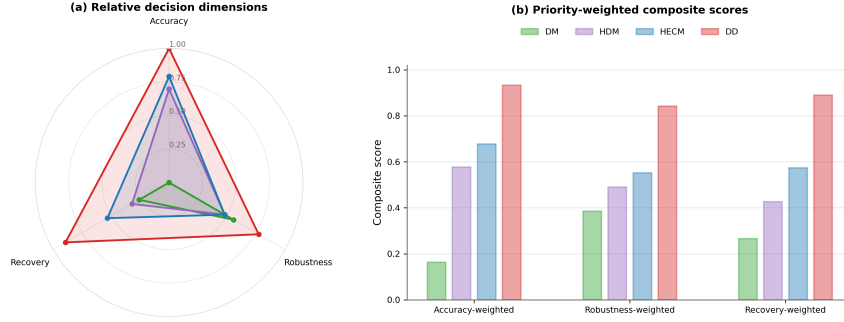


Figure 15: Decision-oriented synthesis. (a) Relative min-max-normalized accuracy, robustness, and recovery dimensions. (b) Composite scores under three illustrative 60/20/20 priority profiles. Scores summarize the present benchmark only and do not replace scenario-level evidence.

bound on peak runtime memory.

Figure 20 makes the DD deployment trade-off explicit. Exact rolling-window inference preserves the trained software semantics but requires roughly 724 ms median inference time, or 72.4% of the one-second period, and 69.8 KiB static RAM. Continuous-state inference reduces latency to approximately 0.425 ms and RAM to 6.5 KiB while retaining similar error on the six nominal sequences. Periodically resetting that state has similar resource use but creates much larger

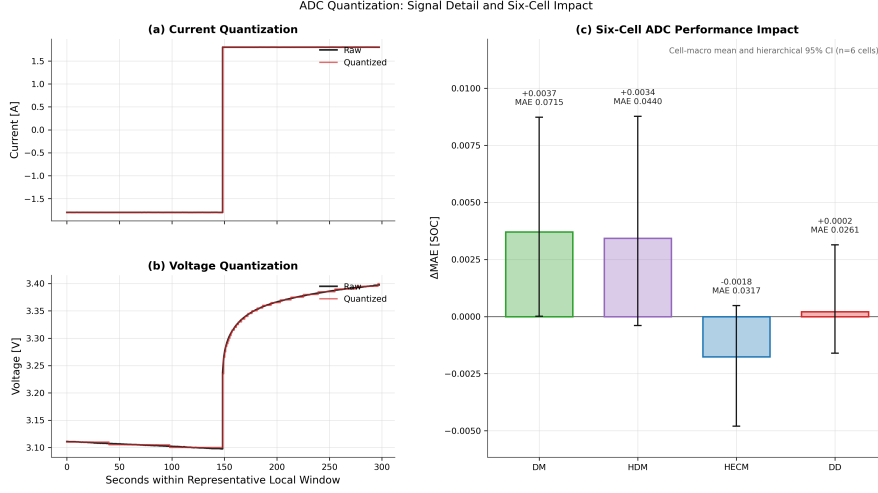


Figure 16: ADC quantization. (a,b) Representative raw and quantized current and voltage. (c) Six-cell Δ MAE with hierarchical 95% confidence intervals and resulting quantized-signal MAE annotations.

transient errors, with a maximum SOC error near 28 percentage points. Continuous state is therefore the practical hardware candidate, but it remains an inference-semantics ablation rather than a drop-in replacement for the primary rolling-window software benchmark.

5. Discussion

5.1. Model-specific performance behavior

DM behaves as expected for a pure integration estimator: it has the largest nominal error and is sensitive to persistent gain error and missing samples because no voltage-feedback correction is available. Full-charge anchoring limits unbounded drift, but the lifecycle experiment shows that it does not erase the accumulated penalty uniformly. HDM changes the capacity denominator through SOH and therefore reduces some aging-related mismatch, yet it retains the same integration mechanism. Its similar current-bias and sample-loss response is consequently structural rather than incidental.

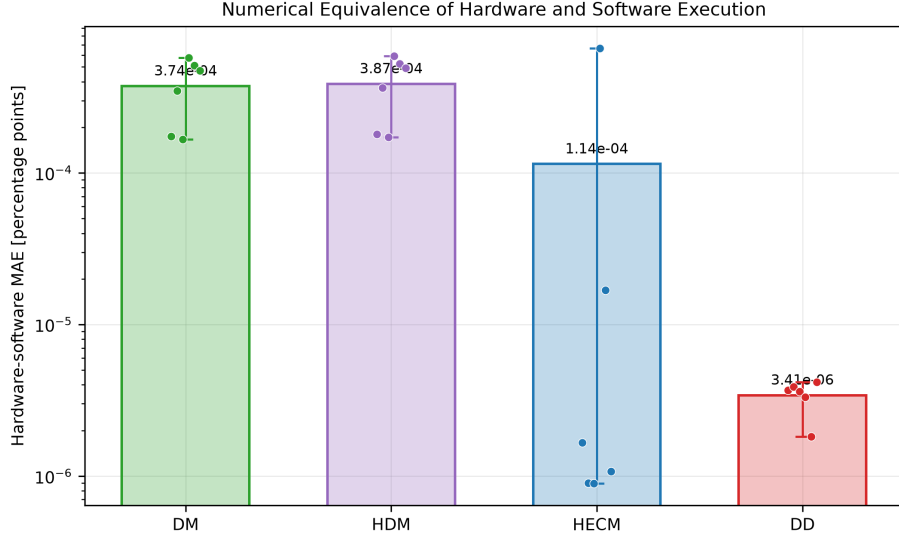


Figure 17: Numerical equivalence of STM32 and software execution over six cell sequences. Bars show mean hardware–software MAE in percentage points, dots show cells, and whiskers show the cell range. The logarithmic scale emphasizes that all implementation differences are negligible relative to model error.

HECM combines current-driven state propagation with a voltage residual and recovers the imposed initial mismatch fastest. This correction capability does not make it universally robust: high current noise and burst dropout produce the largest HECM penalties because both state propagation and observer correction are affected when the causal input stream is corrupted or frozen. The primary campaign does not perturb the fixed HPPC-derived parameter surfaces, so the HECM current-gain result is conditional on this lookup-table quality rather than evidence of equal robustness under parameter misidentification. Its table dependence and flash footprint must therefore be balanced against its rapid initialization correction.

DD has the lowest nominal cell-macro error and the best adverse current-gain score in this comparison. It is also comparatively resilient to isolated sample loss and recovers fastest after burst dropout. Conversely, event-aligned voltage spikes expose a recurrent transient that full-window MAE hides. The result

On-Device Inference-Latency Distributions Across Six Cells

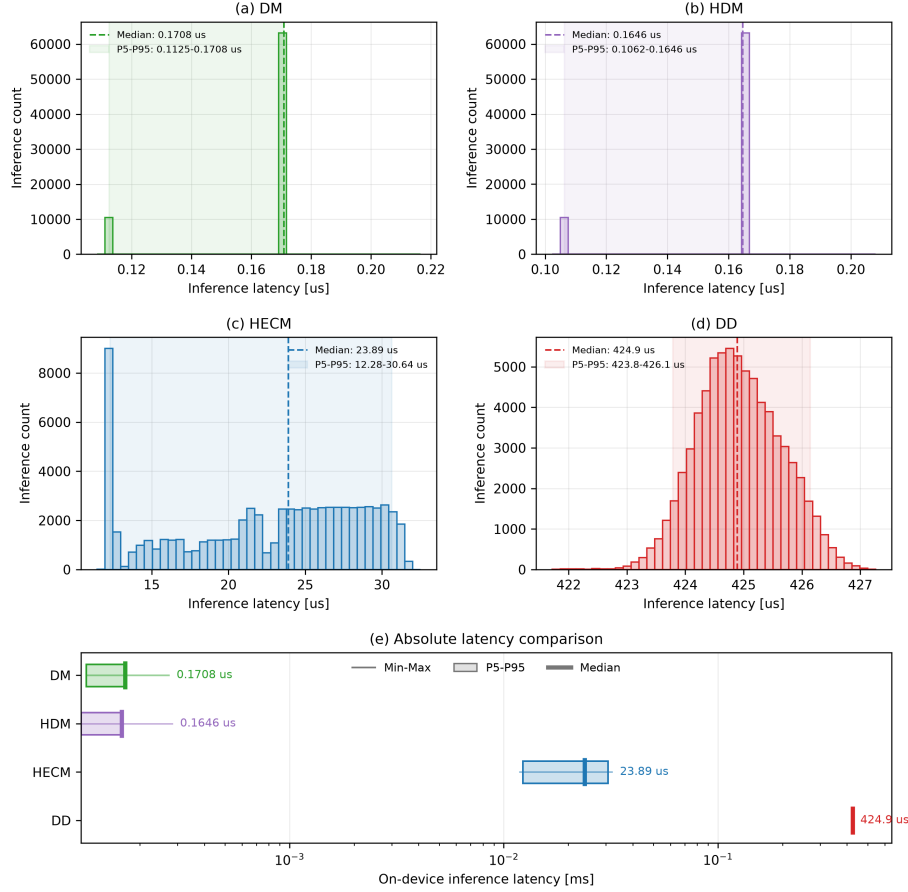


Figure 18: On-device SOC-core latency distributions across six cells and three replay rounds. Panels (a–d) show per-inference distributions; panel (e) compares minimum, 5th percentile, median, 95th percentile, and maximum on a logarithmic scale. DD denotes continuous-state inference in this cross-model latency comparison.

556 is specific to the frozen feature set and checkpoint; it indicates a channel-level
 557 vulnerability of this implementation, not a universal limitation of recurrent SOC
 558 estimators.

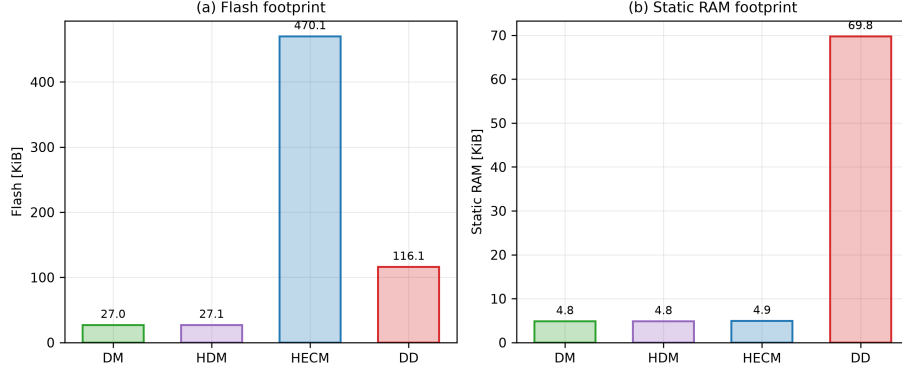


Figure 19: Measured release-image flash and static-RAM footprints for the isolated STM32 SOC implementations. Flash and RAM are firmware properties and are therefore reported once per implementation rather than as artificial cell-dependent distributions.

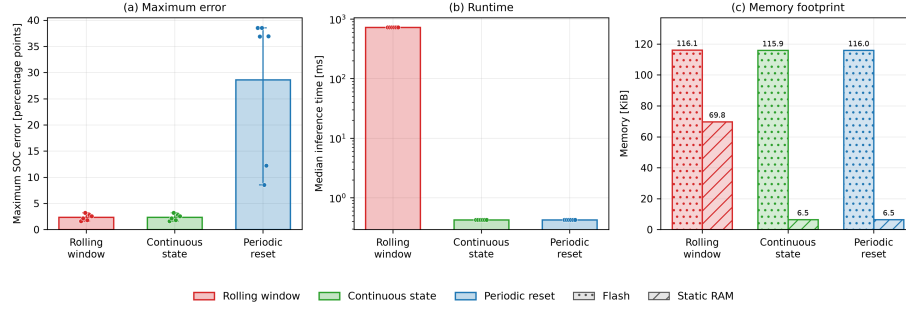


Figure 20: DD inference-mode trade-off on the STM32. The panels compare maximum dataset-SOC error, median latency, and firmware memory for exact rolling-window, continuous-state, and periodically reset recurrent execution. Cell points show that periodic resets, not continuous forwarding, create the dominant accuracy loss.

5.2. Accuracy-performance trade-offs

No single scalar captures the observed trade-offs. DD leads the baseline MAE/RMSE ranking and the three illustrative weighted profiles, but HECM is substantially faster at correcting initialization mismatch and DM/HDM are orders of magnitude cheaper in inference time. Recovery itself is scenario dependent: HECM leads after initialization mismatch, whereas DD leads after burst dropout. Likewise, the reference-SOH ablation improves HDM and HECM but

not DD. Model choice should therefore start from the relevant failure mechanism, latency budget, and state-service architecture rather than from one aggregate rank.

5.3. Implications for BMS deployment

For the tested LFP cells, DD is the strongest general-purpose accuracy/robustness candidate if its recurrent implementation and voltage-spike response are acceptable. HECM remains preferable when restart correction is critical and a larger flash image is available. DM and HDM are attractive when deterministic sub-microsecond execution and small memory dominate, but persistent current calibration and communication continuity then require stronger system-level safeguards. These recommendations apply to the tested models, not to their estimator families in the abstract.

The STM32 results narrow the gap between algorithmic and deployment claims. All four isolated SOC cores meet the one-second deadline with the tested implementation, but their margins differ by more than six orders of magnitude between DM/HDM and exact rolling-window DD. Continuous-state DD removes most of that cost, at the price of changing inference semantics. The hardware test does not include the shared SOH-LSTM, peak stack, energy, pack communication, or concurrent BMS tasks; therefore it supports SOC-core feasibility, not complete BMS schedulability. Table 2 summarizes the evidence without hiding these boundaries.

Figure 15 complements the recommendation map with relative scores, not additional measurements. Lower-is-better inputs are min–max normalized within the four tested implementations. Accuracy combines baseline MAE, RMSE, and 95th-percentile error; robustness combines cross-scenario penalties; and recovery combines initialization and burst-dropout behavior. The three profiles assign 60% to their named dimension and 20% to each remaining dimension. DD scores highest in all three illustrative profiles (0.933, 0.842, and 0.890), followed by HECM, HDM, and DM. This consistency reflects the present input metrics and normalization; it must not be generalized beyond the tested cells

Table 2: Decision-oriented recommendation map derived from the benchmark results. The table summarizes which estimator class is most suitable when one deployment priority dominates the design decision.

Primary deployment priority	Recommended class	Main supporting benchmark evidence	Main trade-off to consider
Lowest nominal SOC error	DD	Lowest six-cell macro MAE (0.0258) and RMSE (0.0300)	Rolling-window execution is the most computationally expensive implementation
Fastest recovery from initialization mismatch	HECM	Mean stable recovery of 1.03 h under the common criterion	Highest flash footprint and slowest post-dropout recovery
Lowest adverse current-gain penalty	DD	Δ MAE 0.0038 at 3%, below HECM, HDM, and DM	Local voltage-spike transient remains visible
Lowest periodic/random sample-loss penalty	DD	Δ MAE 0.0011/0.0026	One-hour burst dropout still increases MAE by 0.0134
Lowest deterministic compute cost	DM/HDM	Sub-microsecond median SOC-core latency and approximately 27 KiB flash	Integration structure is sensitive to gain error and missing charge information
Practical fast neural deployment	DD continuous state	Approximately 0.425 ms median and 6.5 KiB static RAM	Different recurrent semantics from the primary rolling-window benchmark

or interpreted as proof that DD dominates every individual scenario.

5.4. Limitations and future extensions

The study remains limited to six holdout cells from one controlled LFP dataset and therefore cannot establish transfer to NMC/NCA chemistry, other form factors, pack imbalance, other thermal environments, or field duty cycles. The load groups are descriptive strata with $n = 2$ low-load, $n = 3$ middle-load, and one exploratory high-load cell; they are not randomized treatments. C27 contains no mid-life or aged window, and missing states are not imputed. With six independent cells, confidence intervals and effect magnitudes are more informative than dichotomized significance decisions.

Only measurement, timing, availability, and initialization channels are manipulated. Cell/pack parameter mismatch, ECM aging drift, hysteresis error, thermal gradients, sensor cross-sensitivity, balancing and contactor faults, communication-protocol failures, numerical faults, and deadline interference remain outside the campaign. DD initialization is implemented through an SOC-equivalent Q_c offset rather than direct hidden-state corruption. Abrupt capacity-loss events are absent, so the hourly SOH service is evaluated only for

gradual aging. The voltage-spike mechanism is supported by event alignment but not proven by an input-channel ablation. HECM results remain conditional on the imported HPPC lookup tables, which were fixed rather than reidentified for each holdout cell.

The hardware experiment isolates SOC cores and passes SOH as an external causal trace. It does not measure end-to-end SOH latency, peak stack, energy, CPU utilization under concurrent BMS tasks, or pack-level closed-loop behavior. Future work should therefore prioritize peak-memory and energy instrumentation, scheduling tests with the SOH service, clustered-fault experiments, hidden-state ablations, and independent validation on additional chemistries and duty cycles.

6. Conclusion

This work replaces a clean-data, single-trajectory comparison with a causal six-cell LFP benchmark covering 19 measurement-only cases, repeated stochastic seeds, cell-aware uncertainty, and isolated STM32 validation. DD has the lowest nominal macro error and the lowest adverse current-gain penalty in the tested set. HECM corrects initialization mismatch fastest, DD recovers fastest after burst dropout, sample loss matters more than timing jitter, and event alignment exposes a DD voltage-spike transient that global MAE conceals. The paired SOH ablation further shows that reference SOH does not uniformly improve every SOC pipeline.

The practical conclusion is scenario specific. DD provides the strongest aggregate result for these cells but requires careful recurrent deployment semantics and spike-aware validation. HECM offers rapid observer-based re-anchoring at higher flash cost, whereas DM and HDM provide exceptional timing and memory margins with greater dependence on current accuracy and data continuity. The STM32 results confirm close numerical equivalence and ample nominal timing margin for all isolated SOC cores, while leaving complete SOC-SOH scheduling, peak memory, and energy as open system-level validation tasks.

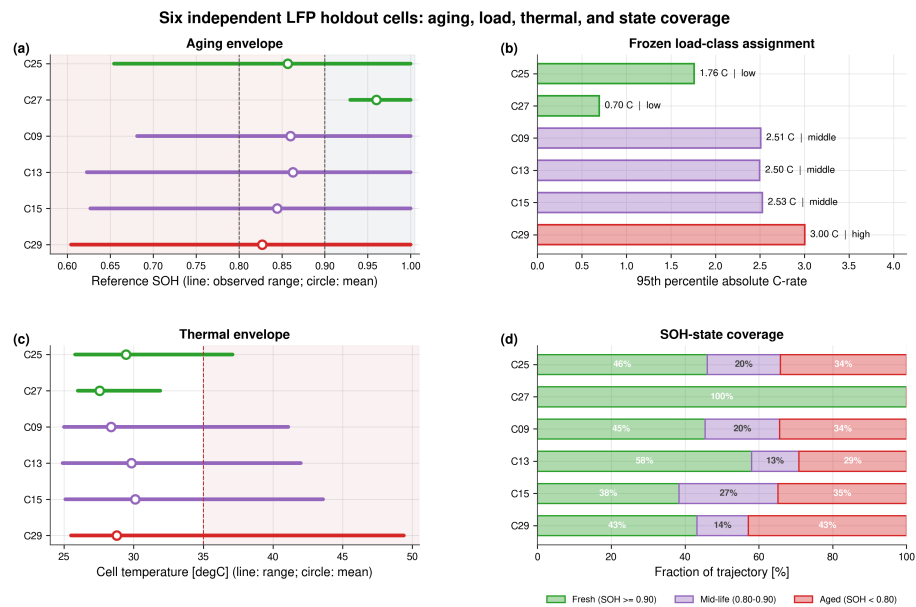


Figure A.21: Quantitative coverage of the six independent holdout cells. Panels show observed SOH, the descriptive load-class assignment, thermal envelope, and full-trajectory SOH-state coverage. High-load evidence consists only of C29 and is exploratory.

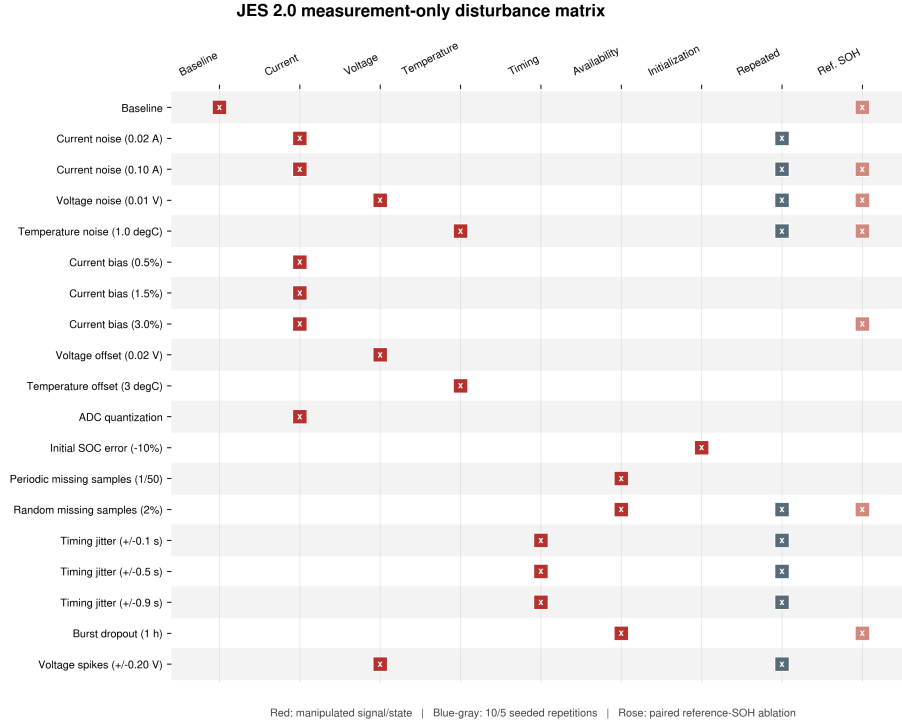


Figure A.22: Complete 19-case JES2 matrix. Red markers identify the manipulated measurement, timing, availability, or initialization channel; blue-gray markers denote repeated stochastic cases; rose markers identify paired reference-SOH ablations.

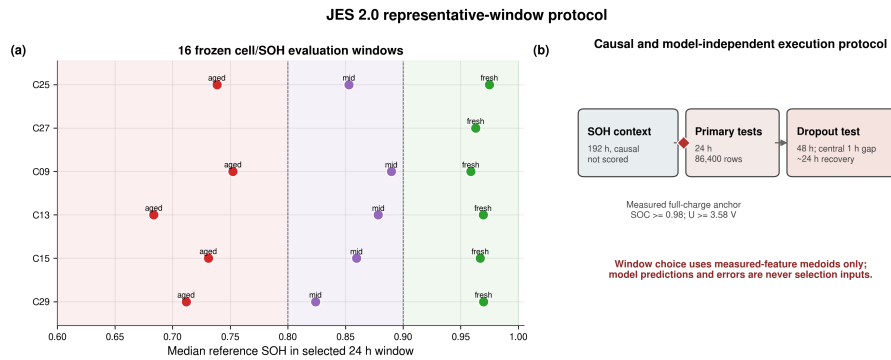


Figure A.23: Frozen representative-window protocol. Sixteen cell/SOH windows are selected using measured-feature medoids only. Each scored 24-h window follows a 192-h causal SOH context; the dropout case extends evaluation to 48 h.

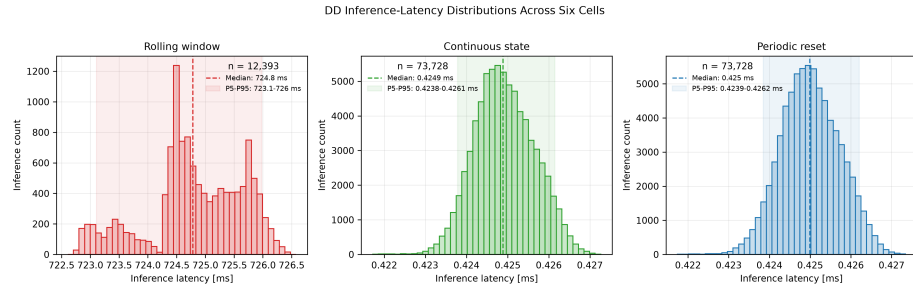


Figure A.24: DD latency distributions over the six hardware cell sequences. Exact rolling-window execution is shown separately from continuous-state and periodically reset execution because their scales and recurrent semantics differ substantially.

Table B.3: Six-cell baseline cell-macro statistics with hierarchical 95% confidence intervals.

Estimator	MAE [SOC]	RMSE [SOC]	P95 error [SOC]
DM	0.0678 [0.0491, 0.0809]	0.0733 [0.0524, 0.0873]	0.1086 [0.0754, 0.1307]
HDM	0.0405 [0.0297, 0.0506]	0.0438 [0.0317, 0.0555]	0.0648 [0.0439, 0.0845]
HECM	0.0335 [0.0246, 0.0429]	0.0389 [0.0289, 0.0496]	0.0646 [0.0467, 0.0812]
DD	0.0258 [0.0189, 0.0323]	0.0300 [0.0224, 0.0367]	0.0508 [0.0375, 0.0618]

Table B.4: Selected cell-macro scenario penalties relative to paired baseline. Current-gain values use the adverse sign from each matched $\pm 3\%$ pair.

Scenario	DM	HDM	HECM	DD
Current noise, $\sigma_I = 0.10$ A	-0.0009	+0.0000	+0.0056	-0.0000
Current gain, adverse 3%	+0.0107	+0.0100	+0.0059	+0.0038
ADC quantization	+0.0037	+0.0034	-0.0018	+0.0002
Random missing samples, 2%	+0.0079	+0.0080	+0.0038	+0.0026
Timing jitter, ± 0.9 s	-0.0003	+0.0003	+0.0006	+0.0001
Burst dropout, 1 h	+0.0067	+0.0134	+0.0221	+0.0134
Voltage spikes, ± 0.20 V	+0.0000	-0.0003	+0.0009	-0.0001

Table B.5: Recovery and target-hardware summary. Recovery values are equal-weight cell means.

Metric	DM	HDM	HECM	DD
Initial stable recovery [h]	22.28	22.28	1.03	5.57
Burst recovery/censor time [h]	5.13	5.67	12.45	4.13
Median SOC-core latency [μ s]	0.171	0.165	23.9	424.9 ^a
Flash [KiB]	27.0	27.1	470.1	116.1
Static RAM [KiB]	4.8	4.8	4.9	69.8

^aContinuous-state DD; exact rolling-window median is approximately 724 ms.

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648 **CRediT authorship contribution statement**

649 The author contribution statement is provided in the non-blinded submission
650 materials.

651 **Declaration of competing interest**

652 The authors declare that they have no known competing financial interests or
653 personal relationships that could have appeared to influence the work reported
654 in this paper.

655 **Declaration of generative AI and AI-assisted technologies**
656 **in the manuscript preparation process**

657 During the preparation of this work the author(s) used ChatGPT, DeepL,
658 and Grammarly in order to improve the linguistic quality and style of the
659 manuscript. After using these tools, the author(s) reviewed and edited the con-
660 tent as needed and take(s) full responsibility for the content of the published
661 article.

662 **Data availability**

663 The dataset is publicly available through its cited TU Berlin repository
664 record [28]; code and results are maintained in a public GitHub repository. A
665 versioned release will archive the fixed manifests and seeds, frozen model/scaler/ECM
666 artifacts and hashes, causal runners, run summaries, analysis and plotting
667 scripts, and environment needed to rebuild all reported results.

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